RAJARATA UNIVERSITY OF SRI LANKA

FACULTY OF APPLIED SCIENCES

DEPARTMENT OF PHYSICAL SCIENCES

MAT 3222 – Numerical Analysis II

Tutorial 02

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1. Given the function: $f(x) = x^3 - 3x$

   I. Estimate $f'(1)$ with h = 0.1 using the forward difference

   II. Estimate $f''(1)$ with h = 0.1 using the forward difference

   III. Calculate the exact values of $f'(1)$ & $f''(1)$ and absolute errors

2. Given the function: $f(x) = 2x^3 - x^2 + 4$

   I. Estimate $f'(2)$ with h = 0.5 using the backward difference

   II. Estimate $f''(2)$ with h = 0.5 using the backward difference

   III. Calculate the exact values of $f'(2)$ & $f''(2)$ and absolute errors

3. Given the function: $f(x) = x^3 - 4x^2 + 5x$

   I. Estimate $f''(2)$ using the second- order forward difference formula with h =0.1

   II. Estimate $f''(2)$ using the second- order backward difference formula with h =0.1

   III. Determine the exact value of $f''(2)$ and state the absolute error for both estimations.

4. Construct first, second and third Forward and backward Difference Table for the function values:

x	*f(x)*
0	0
1	1
2	8
3	27
4	64

5. Use the Trapezoidal Rule to approximate:

   I = $\int_0^2 (x^3 + 1)$ using single subinterval and find the absolute error.

6. Use the Composite Trapezoidal Rule with n=4 subintervals to approximate:

   I = $\int_0^2 (x^3 + 1)$ and find the absolute error.

7. Use Simpson's 1/3 Rule to approximate:

   I = $\int_0^3 (x^2 + 2x)$

8. Use Composite Simpson's 1/3 Rule with n=4 subintervals to approximate:

   I = $\int_0^2 (x^4 - 2x)$

Submit the tutorial on or before 15/09/2026 at the tutor room.